

Sanchit Kabra

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Research Interests: Self-Evolving Agents · Reasoning in Large Language Models · Red Teaming · AI for Scientific Discovery

EDUCATION

Virginia Tech

PhD, MS (exp. 2026) in Computer Science

2024 – Present

Virginia, USA

Birla Institute of Technology and Science, Pilani

Bachelor of Engineering in Computer Science (Honors)

2020 – 2024

Pilani, India

PUBLICATIONS

- [1] **Accelerating Materials Design via LLM-Guided Evolutionary Search**
Sanchit Kabra*, Nikhil Abhyankar*, Saaketh Desai, Chandan K. Reddy
ICLR 2026
- [2] **Decompose, Adapt and Evolve: Towards Efficient Scientific Equation Discovery with Large Language Models**
Sanchit Kabra*, Pouya Behzadifar*, Parshin Shojaee*, Kazem Meidani, Chandan K. Reddy
ICML 2026 · NeurIPS AI for Science Workshop 2025
- [3] **LLM-AutoSciLab: Closed-Loop Scientific Discovery via Active Experimentation with LLMs**
Sanchit Kabra*, Nikhil Abhyankar, Saaketh Desai, Prasad Iyer, Chandan K. Reddy
NeurIPS 2026
- [4] **LLM-ACES: Closed-Loop Discovery of Dynamical Systems with LLM-Guided Adaptive Search**
Sanchit Kabra*, Nikhil Abhyankar*, Sha Li*, Naren Ramakrishnan, Yulia Gel, Chandan K. Reddy
NeurIPS 2026
- [5] **SURFACEBENCH: Can Self-Evolving LLMs Find the Equations of 3D Scientific Surfaces?**
Sanchit Kabra, Shobhnik Kriplani, Parshin Shojaee, Chandan K. Reddy
TMLR 2026

PREPRINTS

- [1] **Reasoning Towards Fairness: Mitigating Bias in Language Models through Reasoning-Guided Fine-Tuning**
Sanchit Kabra, Akshita Jha, Chandan K. Reddy
arXiv 2025
- [2] **RUST-BENCH: Benchmarking LLM Reasoning on Unstructured Text within Structured Tables**
Nikhil Abhyankar, Purvi Chaurasia, **Sanchit Kabra**, Ananya Srivastava, Vivek Gupta, Chandan K. Reddy
arXiv 2025
- [3] **Biased or Flawed? Mitigating Stereotypes in Generative Language Models by Addressing Task-Specific Flaws**
Akshita Jha, **Sanchit Kabra**, Chandan K. Reddy
arXiv 2024

RESEARCH EXPERIENCE

Amazon Science — Applied Science Intern, Stores, Economics and Science

Open-Ended Hypothesis Generation

May 2026 – Present

Palo Alto, CA

- Developing a self recursive hypothesis generation system leveraging customer purchase trajectories to surface latent hypotheses for driving weblab experiments at scale.
- Building an end-to-end auto-research framework to improve demand substitution models along axes of model architecture, feature engineering and hyperparameter optimization.

Virginia Tech — Research Assistant

Advisor: Chandan Reddy

May 2022 – Present

Arlington, VA

RED TEAMING & SECURITY EVALUATION

- **LLMs as Evolutionary Attackers:** Developed a novel LLM-driven evolutionary framework for discovering single-turn jailbreak prompts under strict black-box query budgets.
- Discovered **3× more** unique successful jailbreaks than template-based attacks under the same budget.
- Achieved **100% ASR** and **82–86%** judge-validated ASR on LLaMA-2-7B-Chat and GPT-4o-mini, surfacing structurally distinct attack strategies.

- **Multi-Vulnerability Code Security Benchmarking:** Developed a novel file-level benchmark for multi-vulnerability detection across C, C++, Python, and JavaScript.
- Identified severe performance degradation with recall dropping by **30–50%** and exact-file accuracy decreasing by up to **40%** in dense vulnerability settings.

AI FOR SCIENCE

- **Materials Discovery (LLEMA):** Developed an LLM-guided evolutionary search system for multi-objective materials discovery evaluated across 10 industrial-scale tasks.
- Achieved up to **2× higher hit-rates** than generative and LLM-only baselines, consistently discovering chemically valid and thermodynamically stable candidates.
- Produced stronger Pareto fronts with higher hypervolume and earlier convergence across all tasks.
- **Scientific Equation Discovery (DecAEvolve):** Developed DecAEvolve, an adaptive reinforcement learning based evolutionary framework for LLM-based symbolic regression.
- Reduced discovery error by **1.5–3×** (normalized MSE) vs. the state-of-the-art LLM-SR baseline under identical 3K-query budgets.
- Achieved **1.5–2× faster convergence**, reaching comparable error in **40–60% fewer iterations** across 0.5B and 1.5B models.

RESPONSIBLE AI & ALIGNMENT

- **Bias Analysis and Mitigation in Generative Language Models:** Designed a framework to disentangle identity-conditioned bias from task-specific reasoning failures in ambiguous language understanding.
- Found that **>80%** of stereotypical errors arise from reasoning failures; reduced stereotypical outputs by **60%+** through instruction tuning without fairness-specific data.
- **Reasoning-Guided Bias Mitigation:** Developed a reasoning-guided fine-tuning approach to distill structured reasoning traces into non-reasoning models, improving fairness accuracy from **~40–50%** to **~85–90%**.

PROJECTS

Amazon Trusted AI Challenge — Code-Based LLMs
Advisor: Dr. Ruoxi Jia

October 2024 – June 2025
Amazon / Virginia Tech

- Developed an 8B-parameter code-focused LLM from scratch (pretraining and finetuning) as part of one of 5 blue teams in a global responsible AI competition.
- Engineered a Chain-of-Thought knowledge distillation framework to transfer reasoning from GPT-4-O1 and DeepSeek R1 into an 8B base model.
- Designed adversarial finetuning pipelines to reduce susceptibility to generating malicious or vulnerable code while maintaining functional correctness.
- Evaluated on security benchmarks using static analysis tools and dynamic LLM-driven vulnerability assessment systems.

AWARDS

Amazon Trusted AI Challenge, 2024
Part of **HokieTokie**, one of 10 teams selected globally; received **USD 900,000** in AWS compute credits.

Advisor: Ruoxi Jia

Higher Secondary School Exams State Topper, 2020
Ranked 2nd in Maharashtra State Board among 1.5 million students; scored 100/100 in Math, Chemistry, Physics, and Biology.

INSPIRE Scholarship, 2020
Awarded for finishing in the top 1% in Class XII Board Examinations.

DST, Government of India

SKILLS

Languages: Python, C/C++, Java, R, Matlab, SQL, bash

Frameworks: PyTorch, TensorFlow, HuggingFace, Numpy, Pandas, SciKit-Learn, Matplotlib, pyspark

Technologies: AWS, Linux

RELEVANT COURSEWORK

- Data Structures & Algorithms
 - Design & Analysis of Algorithms
 - Operating Systems
 - Database Systems
- Compiler Construction
 - Computer Architecture
 - Deep Learning
 - Natural Language Processing
- Multimodal Vision
 - Decision Making Under Uncertainty
 - Computational Learning Theory